

2 Model

2.1 Formal Model

We will go over the description of the formal model for the Blue-Eyed Islanders riddle. According to Van Ditmarsch & Van Der Hoek et al. (2007), the model should be an S5 Kripke model in the public announcement logic language $\mathcal{L}_{K\Box}$. We can define the model as $M = \langle S, \sim, V \rangle$ with agents in $A = \{a_0, a_1, a_2, \dots, a_{n-1}\}$, depending on the configuration. In this definition, S is the non-empty set of states, $\sim$ is the set of relations $\sim_i$ for each agent $i \in A$, and V is the valuation function that for each atomic preposition p provides the set of worlds in which it is true ($P \rightarrow \mathcal{P}(S)$).

Each agent represents one islander and has an eye colour. This colour is blue or brown (formally not blue). This can be represented by introducing the propositional atom b_i for each agent i , expressing that agent $i \in A$ has blue eyes.

A world $w \in S$ is represented as a binary vector of length n (number of agents). Moreover, $w[i]$ indicates the eye colour of agent i . $w[i] = 1$ indicates blue eyes and $w[i] = 0$ brown eyes. For example, considering a setting with 3 islanders, the possible world $w = 001$ represents the situation where only agent a_2 has blue eyes. S contains all combinations of eye colour assignments. Therefore: $S = \{0, 1\}^n$, where n is the number of agents.

Relation $\sim_i$ represents which worlds agent i considers possible. In the original description of the riddle, all agents are certain about the eye colour of other agents, and thus the only possible uncertainty could be caused by the variability of their own eye colour. Therefore, agent i cannot distinguish between two worlds s and t iff they agree on the eye colour of all other agents except their own. In other words, $s \sim_i t$ holds iff $s[j] = t[j]$ for all agents j excluding i ($A \setminus \{i\}$), while $s[i]$ may differ from $t[i]$. As mentioned before, the described model M is an S5 Kripke model, meaning that each relation $\sim_i$ is reflexive, symmetric and transitive. This is also in line with the description of the agents that they are perfect logicians. Figure 1 illustrates an example of a resulting Kripke model for 3 agents.

The valuation function V should provide the set of worlds where the atomic proposition b_i is true. Therefore, for all $w \in S$ and $i \in A$: $w \in V(b_i)$ iff $w[i] = 1$.

Public announcements transform a model by restricting it to the worlds where the announcement holds. In PAL, announcements are always truthful and are available to all agents. In the Blue-Eyed Islanders riddle, the initial public announcement “At least one of you has blue eyes.”, can be written in $\mathcal{L}_{K\Box}$ language as “ $b_0 \vee b_1 \vee \dots \vee b_{n-1}$ ”. The next day, in the case that no islander leaves, the inaction can be modeled as the public announcement “ $\neg K_0 b_0 \wedge \neg K_1 b_1 \wedge \dots \wedge \neg K_{n-1} b_{n-1}$ ”. This translates to the sentence “ a_0 does not know if it has blue eyes, and a_1 does not know if it has blue eyes, and ..., and a_{n-1} does not know if it has blue eyes.”. In the original riddle, this will continue until the k th dawn, when all k blue-eyed islanders simultaneously leave the island. This will convey the public announcement “ $K_i b_i \wedge K_j b_j \wedge \dots$ ”, or “ a_i knows it has blue eyes, and a_j knows it has blue eyes, etc.”.

2.2 Visibility Matrices

Our goal is to generalize the Blue-Eyed Islanders riddle by generalizing the perfect visibility assumption. To this end, we introduce visibility matrices E that represent the visibility configurations between agents. Introducing these allows us to examine the varying configurations on the same underlying framework previously described.

We can define the visibility matrix E as a binary matrix $E \in \{0, 1\}^{n \times n}$, where n is the number of agents. The rows and columns describe the different agents, while the entries $E[i][j]$ denote whether agent j (column agent) can see agent i 's eye colour (row agent). $E[i][j] = 1$

indicates that column agent j can see the eye colour of row agent i , while $E[i][j] = 0$ indicates that column agent j cannot see row agent i 's eye colour. Defining E like this allows visibility to be asymmetric. In other words, agent j might see agent i 's eye colour while agent i cannot see j 's. In terms of the riddle, this could be described as some islanders having mirrored glasses. The

visibility matrix for any vanilla Blue-Eyed Islanders riddle looks like $E = \begin{bmatrix} 0 & 1 & 1 & 1 & \dots \\ 1 & 0 & 1 & 1 & \dots \\ 1 & 1 & 0 & 1 & \dots \\ 1 & 1 & 1 & 0 & \dots \\ \dots & \dots & \dots & \dots & 0 \end{bmatrix}$.

For any binary matrix of $n \times n$, there are $2^{n \cdot n}$ different configurations. Keeping the diagonal entries 0 would mean that there are $2^{n \cdot (n-1)}$ distinct matrices. These matrices represent all possible visibility configurations between agents. Consequently, our implementation will allow us to explore how the epistemic dynamics are affected by the (imperfect) visibility configurations.

2.3 From Visibility Matrices to Kripke Models

In order to use the visibility matrices meaningfully for the implementation, we have to define a method to translate them into Kripke models. This will mainly revolve around inducing the accessibility relations in the Kripke models. The new models can then be evaluated and checked for solvability.

The accessibility relations of the formalized model M were described for the original riddle. In this definition, agent i would not be able to distinguish between worlds iff it was certain about the eye colour of others but not its own. In general, the distinguishability between two worlds by an agent is determined by which aspects of the world they can (or cannot) observe. Let E be a visibility matrix and $M = \langle S, \sim, V \rangle$ be an S5 Kripke model, both as described in the previous subsections. We can define O_i as the set of agents which are observable to agent i such that $O_i = \{j \in A \mid E[j][i] = 1\}$. Using this, we can redefine the accessibility relations $\sim_i$ for two worlds s and t as follows: Agent i cannot distinguish between worlds s and t iff they agree on all the eye colours of the agents in O_i . In other words, $s \sim_i t$ holds iff $s[j] = t[j]$ for all $j \in O_i$, while the eye colours of agents outside O_i may differ.

This definition generalizes the model of the classic riddle, where it was only the case that the agent was uncertain about its own eye colour. The definition does not remove the S5 properties from the model, since every relation is still reflexive, symmetric and transitive. Moreover, each visibility matrix has the same underlying set of worlds and valuation function, however it has unique accessibility relations based on the observation structure. This shows the generalizability of our formalization. Finally, we can now analyse how the distinct configurations evolve through the public announcement logic of the riddle and how they affect the agents' ability to draw conclusions about their own states.

2.4 Restrictions

The number of states and the number of visibility matrices can explode when increasing the number of agents in the riddle. For $n = 4$, there would already be 65,536 visibility matrices that would have to be evaluated, opposed to the 512 matrices for $n = 3$. This gives an idea of how quickly the matrices can scale. In the Demo, the user can choose the amount of agents themselves.

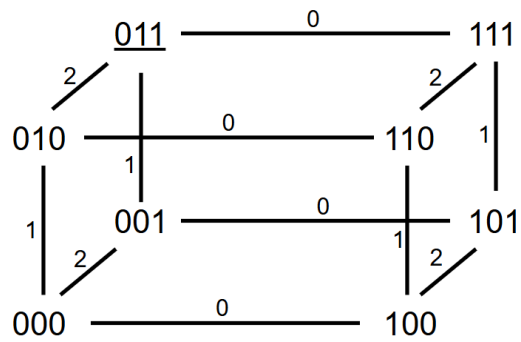


Figure 1: Formal Kripke model for an original Blue-Eyed Islander riddle with 3 agents. The actual world is underlined. Reflexive and symmetric relations are not drawn but are present.